OBSERVABILITY

5 AI Observability Platforms for Multi-Agent Debugging

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TL;DR

Multi-agent systems present unique debugging challenges that traditional monitoring tools cannot address. This guide examines five leading AI observability platforms built for multi-

agent debugging: **Maxim AI** (end-to-end simulation, evaluation, and observability platform), **Arize** (enterprise ML observability with OTEL-based tracing), **Langfuse** (open-source LLM engineering platform), **Braintrust** (evaluation-first platform with purpose-built database), and **LangSmith** (observability for LangChain-based agents). Each platform offers distinct capabilities for tracking agent interactions, debugging complex workflows, and ensuring production reliability.

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Introduction

Multi-agent AI systems have become the backbone of enterprise automation, from autonomous customer support to complex business process orchestration. Yet deploying these systems in production introduces a critical challenge: how do you understand what's happening inside a network of AI agents making autonomous decisions?

Traditional application monitoring tools track uptime and latency, but they cannot answer the questions that matter for multi-agent systems. Which agent made the wrong decision? Why did the workflow fail at step three? How do agents collaborate, and where do handoffs break down? These questions require specialized observability built for the unique architecture of multi-agent AI.

According to IBM's research on AI agent observability, multi-agent systems create unpredictable behavior through complex interactions between autonomous agents. Traditional monitoring falls short because it cannot trace the reasoning paths, tool usage, and inter-agent communication that define how multi-agent systems actually work.

Microsoft's Agent Framework emphasizes that observability has become essential for multi-agent orchestration, with contributions to OpenTelemetry helping standardize tracing and telemetry for agentic systems. This standardization gives teams deeper visibility into agent workflows, tool call invocations, and collaboration patterns critical for debugging and optimization.

In 2026, the AI observability landscape offers several specialized platforms designed to solve these challenges. This guide examines five leading solutions, with particular attention to how they handle the complexities of multi-agent debugging.

Why Multi-Agent Debugging Needs Specialized Observability

Multi-agent systems differ fundamentally from single-agent or traditional software applications. Understanding these differences clarifies why specialized observability matters.

The Multi-Agent Complexity Challenge

Multi-agent systems involve multiple autonomous AI agents working together to complete complex tasks. These agents might handle different aspects of a workflow (such as research, analysis, and execution) or coordinate across specialized domains (like sales pipeline automation with agents for lead qualification, outreach, and scheduling).

Unlike single-agent systems where failures can often be traced to a specific component, multi-agent systems create emergent behaviors through agent interactions. A booking might fail at any point in a travel system with separate agents for flights, hotels, and car rentals. Agent tracing for multi-agent AI systems becomes essential to identify exactly where and why failures occur.

What Makes Multi-Agent Debugging Different

Traditional debugging focuses on deterministic code paths. Multi-agent debugging must account for:

Non-deterministic reasoning: LLMs vary run-to-run, making reproducibility challenging. An agent might make different decisions with identical inputs.

Multi-step tool usage: Agents chain together multiple tool calls (database queries, API requests, web searches) to accomplish tasks. Modern observability platforms must capture these sequences, making the agent's entire workflow transparent.

Inter-agent communication: Agents pass context, intermediate results, and instructions between each other. Observability must trace these handoffs to understand workflow breakdowns.

State management across turns: Multi-turn conversations require tracking how state evolves across agent interactions, including what information each agent has access to.

Quality degradation over time: Unlike code bugs that fail immediately, AI agents can slowly drift in quality. Observability must detect subtle performance changes before they compound.

AI reliability depends on understanding these dynamics across the full agent lifecycle, from development to production deployment.

Key Observability Requirements

Effective multi-agent observability platforms must provide:

- **Distributed tracing** for tracking requests across multiple agents and services
- **Tool call visibility** to see which external functions agents invoke and their results
- **Session-level tracking** for multi-turn conversations and long-running workflows
- **Evaluation integration** to measure quality beyond technical metrics
- **Real-time monitoring** with alerting for production issues
- **Root cause analysis** to quickly identify failure sources in complex agent chains

With these requirements in mind, let's examine five platforms built to address multi-agent debugging challenges.

Platform 1: Maxim AI - End-to-End Agent Observability with Simulation and Evaluation

Maxim AI takes a comprehensive approach to multi-agent observability by integrating simulation, evaluation, and real-time monitoring into a unified platform. This end-to-end philosophy recognizes that production observability alone is not enough; teams need to test and evaluate multi-agent systems before deployment and continuously improve them based on production data.

Platform Overview

Maxim AI is an end-to-end AI simulation, evaluation, and observability platform helping teams ship AI agents reliably and more than 5x faster. The platform serves AI engineers, product managers, and QA teams building multi-agent applications across industries.

What distinguishes Maxim is its unified approach to the AI lifecycle. While many observability tools focus solely on production monitoring, Maxim connects pre-production testing (through agent simulation and evaluation) with production observability. This creates a continuous feedback loop: production data informs better simulations, which improve pre-release testing, resulting in more reliable deployments.

Key Features for Multi-Agent Debugging

Comprehensive Distributed Tracing

Maxim's observability platform captures complete execution traces for multi-agent systems with support for traces, spans, generations, retrieval, tool calls, events, sessions, tags, metadata, and errors. This granularity enables quick debugging and anomaly detection.

For multi-agent systems, tracing captures the full workflow including:

- Agent-to-agent handoffs and context passing
- Tool invocations at each step with inputs and outputs
- LLM calls with prompts, completions, and token usage
- State transitions across the agent workflow
- Errors and their propagation through the agent chain

The platform's tracing concepts documentation details how to instrument multi-agent systems for maximum visibility.

Agent Simulation at Scale

Maxim's unique agent simulation capability allows teams to test multi-agent systems across thousands of real-world scenarios and user personas before production deployment. Simulations capture detailed traces across tools, LLM calls, and state transitions, identifying failure modes early.

For multi-agent debugging, simulation provides:

- Pre-production testing of agent collaboration patterns
- Identification of edge cases in agent handoffs
- Validation of tool usage sequences
- Stress testing under various conditions and personas
- Reproducible test scenarios for regression prevention

Flexi Evaluations for Multi-Agent Systems

Maxim's evaluation framework allows teams to configure evaluations with fine-grained flexibility. While SDKs support running evals at any level of granularity (trace, span, or session), the UI empowers product teams to manage evaluations without writing code.

For multi-agent systems, this means:

- Session-level evaluations for multi-turn conversations
- Trace-level evaluation of complete workflows
- Span-level assessment of individual agent actions
- Custom evaluators (deterministic, statistical, and LLM-as-judge)
- Human-in-the-loop evaluations for nuanced quality checks

Evaluation workflows for AI agents shows how teams structure continuous evaluation processes.

Online Evaluations with Alerting

Production quality monitoring through online evaluations enables continuous scoring of real user interactions. This surfaces regressions early with automated alerting for targeted remediation.

Multi-agent systems benefit from:

- Real-time quality scoring in production
- Threshold-based alerts for degradation
- Trend analysis across custom dimensions
- Automated regression detection
- Integration with incident response workflows

Data Engine for Continuous Improvement

Maxim's data curation capabilities support multi-modal datasets with workflows for:

- Curating high-quality examples from production logs
- Creating targeted evaluation datasets
- Enriching data through human review
- Building simulation scenarios from real interactions

- Continuous evolution using production insights

This creates a virtuous cycle: production data improves test datasets, which improve pre-release validation, which improves production quality.

Custom Dashboards and Saved Views

Teams need deep insights across agent behavior that cut across custom dimensions. Maxim's custom dashboards provide control to create these insights with a few clicks, while saved views enable repeatable debugging workflows across teams.

Cross-Functional Collaboration

Maxim's UX is designed for how AI engineering and product teams collaborate. While the platform provides highly performant SDKs in Python, TypeScript, Java, and Go, the entire experience allows product teams to drive the AI lifecycle without core engineering dependence.

Enterprise-Grade Infrastructure

Maxim supports enterprise deployments with:

- OTLP ingestion and forwarding to external collectors (Snowflake, New Relic, OTEL)
- AI-specific semantic conventions for standardized instrumentation
- Hybrid and self-hosted deployment options for data sovereignty
- SSO integration and role-based access control
- Comprehensive APIs for integration with existing workflows

Best For

Maxim AI is ideal for:

- **Teams building complex multi-agent systems** requiring full-stack simulation, evaluation, and observability

- **Cross-functional teams** where product managers and engineers need shared visibility
- **Organizations prioritizing quality** with emphasis on pre-production testing and continuous improvement
- **Enterprise deployments** requiring data governance, security, and compliance controls
- **Fast-moving teams** that need to ship reliable AI agents 5x faster

Compare Maxim with other platforms to understand specific differentiators for your use case.

Platform 2: Arize - Enterprise ML Observability for Multi-Agent Systems

Arize brings enterprise-grade ML observability expertise to the LLM and AI agent space. The platform serves global enterprises including Handshake, Tripadvisor, PepsiCo, and Microsoft, offering both Arize AX (enterprise solution) and Arize Phoenix (open-source offering).

Platform Overview

Arize secured $70 million in Series C funding in February 2025, demonstrating strong market validation for comprehensive observability capabilities. The platform extends its traditional ML monitoring strengths (drift detection, bias monitoring, embedding analysis) into the LLM and multi-agent domain.

Key Features

- **OTEL-Based Tracing**: OpenTelemetry standards provide framework-agnostic observability with vendor-neutral instrumentation
- **Comprehensive Evaluations**: Robust evaluation tools including LLM-as-a-Judge, human-in-the-loop workflows, and pre-built evaluators
- **Enterprise Monitoring**: Production monitoring with real-time tracking, drift detection, and customizable dashboards
- **Multi-Modal Support**: Unified visibility across traditional ML, computer vision, LLM applications, and multi-agent systems

- **Advanced Drift Detection**: Monitors semantic patterns in model outputs to detect subtle quality changes over time

Best For

Arize excels for:

- Large enterprises with existing MLOps infrastructure
- Teams running both traditional ML and LLM workloads
- Organizations requiring explainability and bias detection
- Regulated industries (finance, healthcare) with compliance requirements

Compare Arize with Maxim AI for detailed feature analysis.

Platform 3: Langfuse - Open-Source LLM Engineering Platform

Langfuse provides an open-source observability platform tailored for LLM applications and agents. By late 2025, Langfuse has gained significant traction with thousands of developers and a growing user base.

Platform Overview

Langfuse strikes a balance between essential functionality and flexibility. Its open-source foundation ensures transparency and allows teams to self-host completely when requirements demand it. The platform offers managed services for teams that prefer cloud hosting while maintaining enterprise features without sacrificing openness.

Key Features

- **LLM Call Tracing & Logging**: Captures detailed traces of LLM calls, including prompts and responses, naturally handling sequences of calls
- **Session Tracking**: Groups related interactions for comprehensive conversation analysis

- **Cost Analytics**: Monitors token usage and tracks expenses across different models and deployments
- **Framework Integration**: Native support for LangGraph, LlamaIndex, OpenAI Agents SDK, and other popular frameworks
- **Self-Hosting Options**: Full control over data with self-hosted deployment capabilities

Best For

Langfuse fits well for:

- Open-source advocates prioritizing transparency and customizability
- Teams with strict data governance requiring self-hosted solutions
- Organizations building custom LLMOps pipelines needing full-stack control
- Budget-conscious startups seeking powerful capabilities without vendor lock-in

Compare Langfuse with Maxim AI to evaluate which approach matches your requirements.

Platform 4: Braintrust - Evaluation-First AI Observability

Braintrust treats production data as the source of truth for quality improvement. The platform features Brainstore, a purpose-built database for AI application logs enabling 80x faster queries compared to traditional databases.

Platform Overview

Braintrust emphasizes systematic evaluation workflows integrating directly into CI/CD pipelines. The platform is used by teams at Notion, Stripe, Zapier, Vercel, Airtable, and Instacart, indicating strong traction in production environments.

Key Features

- **Brainstore Database**: Purpose-built for AI workflows, handling complex telemetry data 80x faster than traditional databases

- **Automated Scoring**: LLM-specific evaluation metrics assessing response quality through semantic understanding
- **CI/CD Integration**: Native GitHub Actions and CircleCI support for quality gates in deployment pipelines
- **Loop AI Agent**: Automated evaluation creation building prompts, datasets, and scorers
- **Production Trace Conversion**: One-click conversion of production failures into evaluation datasets
- **Framework Support**: Native support for 13+ frameworks including LangChain, LlamaIndex, Vercel AI SDK, OpenAI Agents SDK, and more

Best For

Braintrust works well for:

- Teams prioritizing automated evaluation in development workflows
- Organizations needing fast query performance on large trace datasets
- Development teams using CI/CD pipelines for AI deployment
- Companies valuing systematic quality improvement processes

Compare Braintrust with Maxim AI for a detailed analysis.

Platform 5: LangSmith - Observability for LangChain Agents

LangSmith is the observability and evaluation platform offered by the team behind LangChain, one of the most popular frameworks for building AI agents. If your agents are built using LangChain or LangGraph, LangSmith provides tailor-made monitoring.

Platform Overview

Introduced in mid-2023, LangSmith has evolved significantly through 2025. The platform provides a hosted solution for tracing, logging, and evaluating LLM applications, deeply integrated with LangChain's concepts of chains and agents. Its core philosophy is making it

simple for developers to instrument their code and get useful insights during development and after deployment.

Key Features

- **Seamless LangChain Integration**: With minimal code changes (often just environment variables), teams get full visibility into all LangChain operations
- **Detailed Trace Visualization**: See each agent execution as a trace with nested calls, tool invocations, and LLM responses
- **Real-Time Monitoring**: Track business-critical metrics like costs, latency, and response quality with live dashboards and alerts
- **Conversation Clustering**: See clusters of similar conversations to understand user needs and identify systemic issues
- **Development to Production**: Uses the same tracing infrastructure from prototype through production deployment
- **Framework Agnostic**: While optimized for LangChain, works with other frameworks through APIs and SDKs

Best For

LangSmith excels for:

- Developers building with LangChain or LangGraph
- Teams wanting lightweight production monitoring without heavy infrastructure
- Rapid prototyping and debugging in development
- Startups prioritizing ease of setup and frictionless integration

Compare LangSmith with Maxim AI for detailed capabilities comparison.

Platform Comparison at a Glance

FEATURE	MAXIM AI	ARIZE
Core Strength	End-to-end simulation, evaluation, observability	Enterprise ML + I

FEATURE	MAXIM AI	ARIZE
Agent Simulation	✓ Advanced	×
Distributed Tracing	✓ Comprehensive	✓ OTEL-based
Evaluation Framework	✓ Flexi evals (trace/span/session)	✓ Robust
Online Evaluations	✓ With alerting	✓ Yes
Cross-Functional UX	✓ Code + no-code workflows	Engineering-focu
Custom Dashboards	✓ Flexible	✓ Yes
Multi-Modal Support	✓ Full	✓ Yes
Self-Hosting	✓ Enterprise option	✓ Enterprise
Framework Support	All major frameworks	Framework-agno
Data Curation	✓ Advanced workflows	Limited
Pricing Model	Tiered plans	Enterprise custor

How to Choose the Right Platform

Selecting an AI observability platform for multi-agent debugging depends on several factors:

1. Development Lifecycle Needs

Choose Maxim AI if: You need full-stack capabilities spanning simulation, evaluation, and observability. Maxim's integrated approach accelerates teams by connecting pre-production testing with production monitoring.

Choose other platforms if: You only need production observability without simulation or extensive evaluation workflows.

2. Team Structure and Collaboration

Choose Maxim AI if: Cross-functional teams (engineering + product) need shared visibility and workflows. Maxim's no-code capabilities reduce engineering bottlenecks.

Choose other platforms if: Only engineering teams will interact with the observability system.

3. Framework and Technology Stack

Choose LangSmith if: Your agents are built primarily with LangChain or LangGraph and you want seamless integration.

Choose Arize if: You run both traditional ML and LLM workloads requiring unified monitoring.

Choose Langfuse if: You prefer open-source solutions with self-hosting capabilities.

Choose Braintrust if: Evaluation-driven development and CI/CD integration are priorities.

Choose Maxim AI if: You need framework-agnostic observability supporting all major AI frameworks with a unified platform.

4. Enterprise Requirements

For enterprise deployments requiring data sovereignty, compliance controls, and security certifications, both Maxim AI and Arize offer robust enterprise options. Langfuse provides self-hosting capabilities, while Braintrust offers hybrid deployment options.

5. Budget and Pricing Model

Open-source options (Langfuse, Arize Phoenix) offer free self-hosted deployment. Cloud platforms typically use tiered pricing (Maxim AI, LangSmith) or usage-based models (Braintrust). Enterprise plans provide custom pricing with additional features.

6. Quality Assurance Philosophy

Choose Maxim AI if: You emphasize preventing issues through comprehensive pre-production testing and simulation rather than only catching them in production.

Choose Braintrust if: Automated evaluation in CI/CD pipelines is your primary quality gate.

Choose other platforms if: Post-deployment monitoring and debugging are sufficient.

Conclusion

Multi-agent AI systems represent the future of enterprise automation, but their complexity demands specialized observability. The five platforms examined in this guide each address multi-agent debugging challenges with different philosophies and strengths.

Maxim AI stands out with its end-to-end approach, connecting agent simulation and evaluation with production observability to create a continuous improvement cycle. This comprehensive platform helps cross-functional teams ship reliable AI agents 5x faster while maintaining quality through every stage of the AI lifecycle.

Arize brings enterprise-grade ML monitoring expertise to the LLM space with strong drift detection and compliance capabilities. Langfuse offers open-source flexibility with self-hosting options. Braintrust emphasizes evaluation-first workflows with purpose-built infrastructure. LangSmith provides seamless integration for LangChain-based agents.

The right choice depends on your development lifecycle needs, team structure, framework requirements, and quality assurance philosophy. As AI agent systems become more complex, investing in proper observability becomes not just beneficial but essential for production reliability.

To explore how Maxim AI can accelerate your multi-agent development with comprehensive simulation, evaluation, and observability, schedule a demo or dive into what AI evals are to understand the foundation of quality AI systems.

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Additional Resources

- LLM Observability: How to Monitor Large Language Models in Production
- AI Agent Quality Evaluation
- Agent Evaluation vs Model Evaluation: What's the Difference
- How to Ensure Reliability of AI Applications
- OpenTelemetry: AI Agent Observability Standards
- Microsoft Agent Framework for Multi-Agent Systems

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